

Ralchukwu Mogbogu

Cluster Documents

ANLY 540

8/27/2024

```
##1
```

```
#Install and import pandas.
```

```
import pandas as pd
```

```
import nltk
```

```
import numpy as np
```

```
import string
```

```
import re
```

```
import unicodedata
```

```
import string
```

```
#Import the Jobs.csv file
```

```
from google.colab import files
```

```
uploaded = files.upload()
```



Choose Files Mental-Health-Twitter.csv

- **Mental-Health-Twitter.csv**(text/csv) - 3343511 bytes, last modified: 8/1/2024 - 100% done  
Saving Mental-Health-Twitter.csv to Mental-Health-Twitter.csv

```
mhealth = pd.read_csv('Mental-Health-Twitter.csv')
```

```
##print first 5 rows
```

```
mhealth = mhealth.drop('Unnamed: 0', axis =1)
```

```
mhealth.head()
```



|   | post_id      | post_created                         | post_text                                                           | user_id      | followers | friends | favourites | s |
|---|--------------|--------------------------------------|---------------------------------------------------------------------|--------------|-----------|---------|------------|---|
| 0 | 6.378950e+17 | Sun Aug 30<br>07:48:37<br>+0000 2015 | It's just<br>over 2<br>years<br>since I was<br>diagnosed<br>w...    | 1.013187e+09 | 84        | 211     | 251        |   |
| 1 | 6.378900e+17 | Sun Aug 30<br>07:31:33<br>+0000 2015 | It's<br>Sunday, I<br>need a<br>break, so<br>I'm<br>planning<br>t... | 1.013187e+09 | 84        | 211     | 251        |   |
| 2 | 6.377490e+17 | Sat Aug 29<br>22:11:07<br>+0000 2015 | Awake but<br>tired. I<br>need to<br>sleep but<br>my brain ...       | 1.013187e+09 | 84        | 211     | 251        |   |
| 3 | 6.376960e+17 | Sat Aug 29<br>18:40:49<br>+0000 2015 | RT<br>@SewHQ:<br>#Retro<br>bears<br>make<br>perfect<br>gifts and... | 1.013187e+09 | 84        | 211     | 251        |   |
| 4 | 6.376960e+17 | Sat Aug 29<br>18:40:26<br>+0000 2015 | It's hard to<br>say<br>whether<br>packing<br>lists are<br>mak...    | 1.013187e+09 | 84        | 211     | 251        |   |



Next  
steps:

[Generate code  
with](#) mhealth



[View recommended  
plots](#)

[New interactive  
sheet](#)

```
##post_id
#normalize encodings

#import unicodedata

# Dataframe
mhealth = pd.read_csv('Mental-Health-Twitter.csv', encoding = 'utf8')
mhealth['post_text'] = mhealth['post_text'].apply(lambda x: unicodedata.normalize('NFKD', x))

print(mhealth[0:10])
```



|   | Unnamed: 0 | post_id      | post_created \                 |
|---|------------|--------------|--------------------------------|
| 0 | 0          | 6.378950e+17 | Sun Aug 30 07:48:37 +0000 2015 |
| 1 | 1          | 6.378900e+17 | Sun Aug 30 07:31:33 +0000 2015 |
| 2 | 2          | 6.377490e+17 | Sat Aug 29 22:11:07 +0000 2015 |
| 3 | 3          | 6.376960e+17 | Sat Aug 29 18:40:49 +0000 2015 |
| 4 | 4          | 6.376960e+17 | Sat Aug 29 18:40:26 +0000 2015 |
| 5 | 5          | 6.376930e+17 | Sat Aug 29 18:26:24 +0000 2015 |
| 6 | 6          | 6.376920e+17 | Sat Aug 29 18:21:51 +0000 2015 |
| 7 | 7          | 6.376890e+17 | Sat Aug 29 18:12:59 +0000 2015 |
| 8 | 8          | 6.376870e+17 | Sat Aug 29 18:04:05 +0000 2015 |
| 9 | 9          | 6.376850e+17 | Sat Aug 29 17:54:54 +0000 2015 |

|   | post_text                                         | user_id      | followers \ |
|---|---------------------------------------------------|--------------|-------------|
| 0 | It's just over 2 years since I was diagnosed w... | 1.013187e+09 | 84          |
| 1 | It's Sunday, I need a break, so I'm planning t... | 1.013187e+09 | 84          |
| 2 | Awake but tired. I need to sleep but my brain ... | 1.013187e+09 | 84          |
| 3 | RT @SewHQ: #Retro bears make perfect gifts and... | 1.013187e+09 | 84          |
| 4 | Its hard to say whether packing lists are maki... | 1.013187e+09 | 84          |
| 5 | Making packing lists is my new hobby... #movin... | 1.013187e+09 | 84          |
| 6 | At what point does keeping stuff for nostalgic... | 1.013187e+09 | 84          |
| 7 | Currently in the finding-boxes-of-random-shit ... | 1.013187e+09 | 84          |
| 8 | Can't be bothered to cook, take away on the wa... | 1.013187e+09 | 84          |
| 9 | RT @itventsnews: ITV releases promo video for ... | 1.013187e+09 | 84          |

|   | friends | favourites | statuses | retweets | label |
|---|---------|------------|----------|----------|-------|
| 0 | 211     | 251        | 837      | 0        | 1     |
| 1 | 211     | 251        | 837      | 1        | 1     |
| 2 | 211     | 251        | 837      | 0        | 1     |
| 3 | 211     | 251        | 837      | 2        | 1     |
| 4 | 211     | 251        | 837      | 1        | 1     |
| 5 | 211     | 251        | 837      | 1        | 1     |
| 6 | 211     | 251        | 837      | 1        | 1     |
| 7 | 211     | 251        | 837      | 0        | 1     |
| 8 | 211     | 251        | 837      | 0        | 1     |
| 9 | 211     | 251        | 837      | 41       | 1     |

```
#lowercase texts
```

```
mhealth['post_text']=mhealth['post_text'].str.lower()
```

```
mhealth['post_text'].head()
```

**post\_text**

```
0    it's just over 2 years since i was diagnosed w...
1    it's sunday, i need a break, so i'm planning t...
2    awake but tired. i need to sleep but my brain ...
3    rt @sewhq: #retro bears make perfect gifts and...
4    its hard to say whether packing lists are maki...
```

**dtype:** object

```
#remove punctuations and numbers
mhealth['post_text'] = mhealth['post_text'].str.replace(r'^\w\s','', regex = True)

mhealth['post_text'] = mhealth['post_text'].str.replace(r'\brt\b', '', regex = True, case=F

mhealth['post_text'] = mhealth['post_text'].str.replace('\d+', '', regex = True)
mhealth['post_text'].head(5)
```

**post\_text**

```
0    years since diagnosed anxiety depression today...
1    sunday need break im planning spend little tim...
2           awake tired need sleep brain ideas
3    sewhq retro bears make perfect gifts great beg...
4    hard say whether packing lists making life eas...
```

**dtype:** object

```
#stopwords
nltk.download('stopwords')
from nltk.corpus import stopwords
print(stopwords.words("english"))

['i', 'me', 'my', 'myself', 'we', 'our', 'ours', 'ourselves', 'you', "you're", "you've",
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.
```



```
stop_words=set(stopwords.words("english"))  
stop_words
```

 { 'a',  
 'about',  
 'above',  
 'after',  
 'again',  
 'against',  
 'ain',  
 'all',  
 'am',  
 'an',  
 'and',  
 'any',  
 'are',  
 'aren',  
 "aren't",  
 'as',  
 'at',  
 'be',  
 'because',  
 'been',  
 'before',  
 'being',  
 'below',  
 'between',  
 'both',  
 'but',  
 'by',  
 'can',  
 'couldn',  
 "couldn't",  
 'd',  
 'did',  
 'didn',  
 "didn't",  
 'do',  
 'does',  
 'doesn',  
 "doesn't",  
 'doing',  
 'don',  
 "don't",  
 'down',  
 'during',  
 'each',  
 'few',  
 'for',  
 'from',  
 'further',  
 'had',  
 'hadn',  
 "hadn't",  
 'has',

```
'hasn',
"hasn't",
'have',
'haven',
"haven't",
'having',
```

```
def remove_stop(text): ##I defined it as "text" representing each entry in post_text
    return ' '.join([word for word in text.split() if word not in stop_words])
```

```
mhealth['post_text']=mhealth['post_text'].apply(lambda text : remove_stop(text))
mhealth.head(5)
```

|   | Unnamed: 0 | post_id      | post_created                   | post_text                                         | user_id      | followers | friends | fav |
|---|------------|--------------|--------------------------------|---------------------------------------------------|--------------|-----------|---------|-----|
| 0 | 0          | 6.378950e+17 | Sun Aug 30 07:48:37 +0000 2015 | years since diagnosed anxiety depression today... | 1.013187e+09 | 84        | 211     |     |
| 1 | 1          | 6.378900e+17 | Sun Aug 30 07:31:33 +0000 2015 | sunday need break im planning spend little tim... | 1.013187e+09 | 84        | 211     |     |
| 2 | 2          | 6.377490e+17 | Sat Aug 29 22:11:07 +0000 2015 | awake tired need sleep brain ideas                | 1.013187e+09 | 84        | 211     |     |
| 3 | 3          | 6.376960e+17 | Sat Aug 29 18:40:49 +0000 2015 | sewhq retro bears make perfect gifts great beg... | 1.013187e+09 | 84        | 211     |     |
| 4 | 4          | 6.376960e+17 | Sat Aug 29 18:40:26 +0000 2015 | hard say whether packing lists making life eas... | 1.013187e+09 | 84        | 211     |     |

Next steps:

Generate code with mhealth

View recommended plots

New interactive sheet

```
!pip install spacy
!python -m spacy download en_core_web_sm
```

```
import spacy
nlp = spacy.load('en_core_web_sm')
```

```
Requirement already satisfied: spacy in /usr/local/lib/python3.10/dist-packages (3.8.0)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.10/dist-packages (3.0.11)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.10/dist-packages (1.0.0)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.10/dist-packages (0.28.0)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.10/dist-packages (2.0.2)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.10/dist-packages (3.0.2)
Requirement already satisfied: thinc<8.3.0,>=8.2.2 in /usr/local/lib/python3.10/dist-packages (8.2.2)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.10/dist-packages (0.9.1)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.10/dist-packages (2.4.3)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.10/dist-packages (2.0.6)
Requirement already satisfied: weasel<0.5.0,>=0.1.0 in /usr/local/lib/python3.10/dist-packages (0.1.0)
Requirement already satisfied: typer<1.0.0,>=0.3.0 in /usr/local/lib/python3.10/dist-packages (0.3.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.10/dist-packages (4.38.0)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.10/dist-packages (2.13.0)
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.10/dist-packages (1.7.4)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.10/dist-packages (3.1.2)
Requirement already satisfied: setuptools in /usr/local/lib/python3.10/dist-packages (57.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.10/dist-packages (23.1)
Requirement already satisfied: langcodes<4.0.0,>=3.2.0 in /usr/local/lib/python3.10/dist-packages (3.2.0)
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.10/dist-packages (1.24.2)
Requirement already satisfied: language-data>=1.2 in /usr/local/lib/python3.10/dist-packages (1.2)
Requirement already satisfied: annotated-types>=0.4.0 in /usr/local/lib/python3.10/dist-packages (0.4.0)
Requirement already satisfied: pydantic-core==2.20.1 in /usr/local/lib/python3.10/dist-packages (2.20.1)
Requirement already satisfied: typing-extensions>=4.6.1 in /usr/local/lib/python3.10/dist-packages (4.6.1)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.10/dist-packages (2.0.12)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/dist-packages (2.5)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.10/dist-packages (1.21.1)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.10/dist-packages (2017.4.17)
Requirement already satisfied: blis<0.8.0,>=0.7.8 in /usr/local/lib/python3.10/dist-packages (0.7.8)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.10/dist-packages (0.0.1)
Requirement already satisfied: click>=8.0.0 in /usr/local/lib/python3.10/dist-packages (8.0.0)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.10/dist-packages (1.3.0)
Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.10/dist-packages (10.11.0)
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.10/dist-packages (0.7.0)
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.10/dist-packages (5.2.1)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.10/dist-packages (2.0.1)
Requirement already satisfied: marisa-trie>=0.7.7 in /usr/local/lib/python3.10/dist-packages (0.7.7)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.10/dist-packages (2.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.10/dist-packages (2.13.0)
Requirement already satisfied: wrapt in /usr/local/lib/python3.10/dist-packages (1.12.1)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.10/dist-packages (0.1.0)
Collecting en-core-web-sm==3.7.1
```

Downloading [https://github.com/explosion/spacy-models/releases/download/en\\_core\\_w](https://github.com/explosion/spacy-models/releases/download/en_core_w)  
 12.8/12.8 MB 54.4 MB/s eta 0:00:00

```
Requirement already satisfied: spacy<3.8.0,>=3.7.2 in /usr/local/lib/python3.10/dist-packages (3.7.2)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.10/dist-packages (3.0.11)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.10/dist-packages (1.0.0)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.10/dist-packages (0.28.0)
```

```
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.10/dis
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.10/c
Requirement already satisfied: thinc<8.3.0,>=8.2.2 in /usr/local/lib/python3.10/dis
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.10/di
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.10/dis
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.10
Requirement already satisfied: weasel<0.5.0,>=0.1.0 in /usr/local/lib/python3.10/di
```

```
##Lemmatize
```

```
mhealthX = mhealth.iloc[:1000]
#subset only using first 1000 rows of the df to avoid high memory usage issues-
#from large data
```

```
mhealthX['post_text'] = mhealthX['post_text'].apply(lambda row: " ".join([w.lemma_ for w in
mhealthX['post_text'].head(3)
```

```
##just for the 1st 1000 rows
```



<ipython-input-18-e9e1cfca0555>:1: SettingWithCopyWarning:  
A value is trying to be set on a copy of a slice from a DataFrame.  
Try using .loc[row\_indexer,col\_indexer] = value instead

See the caveats in the documentation: <https://pandas.pydata.org/pandas-docs/stable/user>

```
mhealthX['post_text'] = mhealthX['post_text'].apply(lambda row: " ".join([w.lemma_ for
```

**post\_text**

- 
- 0 year since diagnose anxiety depression today I...
  - 1 sunday need break I m plan spend little time p...
  - 2 awake tired need sleep brain idea

**dtype: object**

```
#just to double-check
mhealthX['post_text'].head(5)
```



**post\_text**

- 0 year since diagnose anxiety depression today I...
- 1 sunday need break I m plan spend little time p...
- 2 awake tired need sleep brain idea
- 3 sewhq retro bear make perfect gift great begin...
- 4 hard say whether pack list make life easy rein...

**dtype:** object

3. Vectorize your text.
4. Reduce the dimensions to two dimensions using PCA or other technique.
5. Cluster your documents using K-means or K-medoids.
6. Use word clouds to label your clusters.
7. Plot the clusters.
8. Compare the cluster labels to at least one meta-data variable.

## Vectorize textdata

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans ##for cluster
import matplotlib.pyplot as plt ##plots
```

```
documentsX = list(mhealthX['post_text'])
vectorizer = TfidfVectorizer()
vectorized_docX = vectorizer.fit_transform(documentsX)
vectorized_docX
```



```
<1000x2587 sparse matrix of type '<class 'numpy.float64'>'
with 5874 stored elements in Compressed Sparse Row format>
```

## Reduce to two dimensions via PCA

```
pca = PCA(n_components=2)
reduced_docX = pca.fit_transform(vectorized_docX.toarray())
```

```
## Cluster documentsX using K-means
```

```
n_clusters = 3 #as shown in class, 3 clusters in this model
kmeans = KMeans(n_clusters=n_clusters, n_init=5,
                 max_iter=500, random_state=42)
kmeans.fit(vectorized_docX)
```



```
KMeans
KMeans(max_iter=500, n_clusters=3, n_init=5, random_state=42)
```

```
## store results in another df
```

```
results = pd.DataFrame()
results['document'] = documentsX
results['cluster'] = kmeans.labels_
```

```
# print the results
print(results.sample(50))
```



|     | document                                           | cluster |
|-----|----------------------------------------------------|---------|
| 150 | beautiful sky tonight nofilter httpcocdirpbq       | 0       |
| 112 | there s either eyelash elephant eye either way...  | 0       |
| 510 | conversion therapy ban obama administration ht...  | 0       |
| 765 | trick help reduce stress httpstconjypizgvtc        | 0       |
| 563 | take sick day you re well mentalhealth httpstco... | 0       |
| 949 | want favorite band know you re special adore y...  | 0       |
| 898 | guy get to show get many follower want like co...  | 0       |
| 310 | stress effect health httpstcogxxcfmgpf             | 1       |
| 9   | itventsnew itv release promo video final serie...  | 0       |
| 99  | love story httpstconjkcbjqspg                      | 0       |
| 587 | child inherit anxiety httpstcoqlskcy               | 2       |
| 476 | campus suicide rate rise httpstcosxdgujjw          | 0       |
| 164 | cambridgenewsuk look around milton country par...  | 0       |
| 24  | much want pizza right                              | 0       |
| 882 | discipline enough practice daily mindfulness h...  | 0       |
| 356 | woman great risk ptsd man httpstcodjsi             | 0       |
| 125 | abigail_titmuss yrs ago puppy abandoned irelan...  | 0       |
| 920 | sagebrush brionnahillxoxo                          | 0       |
| 288 | message mental illness worldwide mentalillness...  | 1       |
| 108 | designtaxi good time post facebook instagram t...  | 0       |
| 155 | gorilla amazing reaction reunite man take care...  | 0       |
| 338 | anyone express concern much worry anxiety http...  | 0       |
| 957 | unhealthy obsession tell well favorite wizard ...  | 0       |
| 147 | rspca_official marley zena dogrescuer look fab...  | 0       |
| 889 | chance anxiety sufferer may smart httpstcocqgzq... | 0       |
| 804 | keep open someone trust                            | 0       |
| 897 | averi_xoxo stalk work                              | 0       |
| 132 | gorgeous day drive ely ely catherdral httpstco...  | 0       |
| 978 | benbuckwalter diagnose depression anxiety admi...  | 0       |
| 461 | employer need offer mental health day httpstco...  | 1       |
| 635 | heroin use rise httpstcoaienljh                    | 0       |

|     |                                                    |   |
|-----|----------------------------------------------------|---|
| 503 | maximize mental health coverage mentalhealth h...  | 1 |
| 465 | plan tackle addiction httpstcorlxpksc              | 0 |
| 850 | significant undertreatment depression httpstcoz... | 0 |
| 472 | help grieve teen httpstcosnhicwph                  | 0 |
| 721 | think habit become addiction                       | 0 |
| 421 | ever want ask someone bisexuality httpstcopsvh...  | 0 |
| 425 | myth age httpstcovdemlpkfu                         | 0 |
| 667 | self pity make other want around relationship      | 0 |
| 271 | sex couple fertility benefit httpstcomddzhdol      | 0 |
| 76  | dusk warm cool httpstcoizfytpbg                    | 0 |
| 604 | celebrity diagnose mentalillness httpstcoswobwhat  | 0 |
| 136 | block another traditional grannysquare crochet...  | 0 |
| 568 | claim hospital bully httpstconhxqjqk               | 0 |
| 162 | square lacy crochet pale blue dk yarn crochet ...  | 0 |
| 625 | schizophrenia possibly combination disorder ht...  | 0 |
| 885 | caffeine addiction look httpstcognmroudtk          | 0 |
| 997 | I m average girl amazing music taste awesome h...  | 0 |
| 238 | mum surgery seem fine overnight hospital home ...  | 0 |
| 734 | addiction find puzzle                              | 0 |

## Word clouds for cluster labeling

```
from wordcloud import WordCloud
```

```
def wordcloud_clusters(model, vectors, features, no_top_words=40):
    for cluster in np.unique(model.labels_):
        size = {}
        words = vectors[model.labels_ == cluster].sum(axis=0).A[0]
        largest = words.argsort()[::-1] # invert sort order
        for i in range(0, no_top_words):
            size[features[largest[i]]] = abs(words[largest[i]])
        wc = WordCloud(background_color="white", max_words=100, width=650, height=200)
        wc.generate_from_frequencies(size)
        plt.figure(figsize=(12,12))
        plt.imshow(wc, interpolation='bilinear')
        plt.axis("off")
        # save model
        plt.savefig(f'cluster{cluster}.png')
        plt.show()
        plt.close()
```

```
wordcloud_clusters(kmeans, vectorized_docX, vectorizer.get_feature_names_out())
```

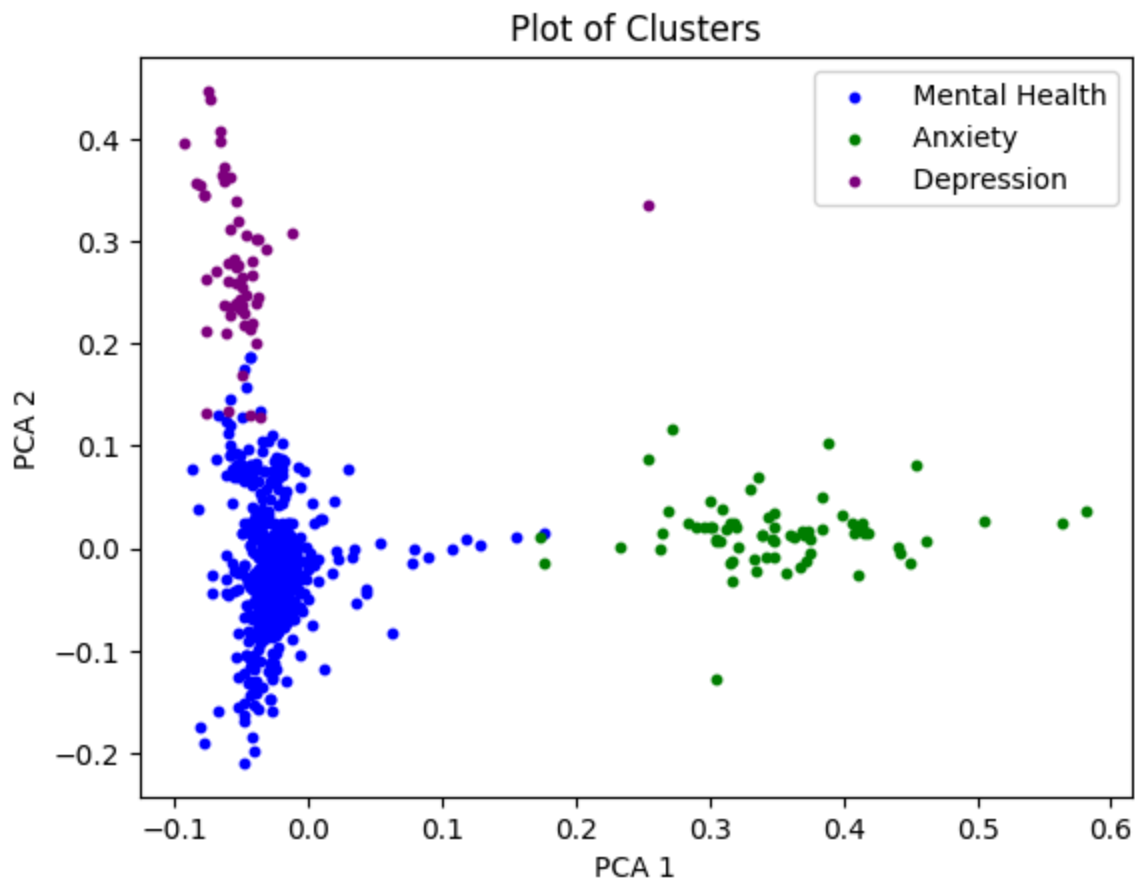


```

colors = ['blue', 'green', 'purple']
cluster = ['Mental Health', 'Anxiety', 'Depression']
for i in range(n_clusters):
    plt.scatter(reduced_docX[kmeans.labels_ == i, 0],
                reduced_docX[kmeans.labels_ == i, 1],
                s=10, color=colors[i],
                label=f' {cluster[i]}')

plt.legend()
plt.xlabel('PCA 1')
plt.ylabel('PCA 2')
##labels are due to reduced dimensions for clustering- "reduced_docX"
plt.title('Plot of Clusters')
##plot title
plt.show()

```



```

## compare cluster labels to a metadata variable;
## metadata: favourites(likes) and retweets
## metadata already numerical

mhealthX['favourite_level'] = 'Mid'
mhealthX.loc[mhealthX['favourites'] > 600, 'favourite_level'] = 'High'
mhealthX.loc[mhealthX['favourites'] == 400, 'favourite_level'] = 'Mid'
mhealthX.loc[mhealthX['favourites'] <= 200, 'favourite_level'] = 'Low'

mhealthX.reset_index(drop=True,inplace = True)

results['favourite_cat'] = mhealthX['favourite_level']

pd.crosstab(results['cluster'], results['favourite_cat'])

```



<ipython-input-73-915f50b93d6c>:5: SettingWithCopyWarning:  
A value is trying to be set on a copy of a slice from a DataFrame.  
Try using .loc[row\_indexer,col\_indexer] = value instead

See the caveats in the documentation: <https://pandas.pydata.org/pandas-docs/stable/user>

```

mhealthX['retweet_level'] = 'Mid'
mhealthX.loc[mhealthX['retweets'] > 500, 'retweet_level'] = 'High'
mhealthX.loc[mhealthX['retweets'] == 200, 'retweet_level'] = 'Mid'
mhealthX.loc[mhealthX['retweets'] <= 100, 'retweet_level'] = 'Low'

mhealthX.reset_index(drop=True,inplace = True)

results['retweet_cat'] = mhealthX['retweet_level']

pd.crosstab(results['cluster'], results['retweet_cat'])

```



<ipython-input-61-ac084ca806cb>:1: SettingWithCopyWarning:  
A value is trying to be set on a copy of a slice from a DataFrame.  
Try using .loc[row\_indexer,col\_indexer] = value instead

See the caveats in the documentation: <https://pandas.pydata.org/pandas-docs/stable/user>

```
mhealthX['retweet_level'] = 'Mid'
```

```
retweet_cat  High  Low  Mid
```

| cluster | High | Low | Mid |
|---------|------|-----|-----|
| 0       | 9    | 860 | 5   |
| 1       | 0    | 70  | 0   |